

Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition

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Instructions

- Focus on the core ideas: MotionEpic (video MLLM) and the

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Related Work

- **MLLMs**: Strong language/image reasoning; video adds temporal complexity.
- **CoT prompting**: Improves step-by-step reasoning; limited adaptation to video.
- **Scene Graphs (SG/STSG)**: Structured semantics for objects, relations, and temporal dynamics; effective for compositional modeling.
- Our contribution: **MotionEpic** integrates STSG grounding within a video MLLM; **VoT** extends CoT to video by chaining perception-to-cognition steps.

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Motivation

- Gap: Complex video reasoning requires both **fine-grained spatiotemporal perception** and **high-level cognitive understanding**.
- Existing video MLLMs: often shallow instance/patch-level analysis; limited commonsense reasoning.
- Our idea: integrate **video spatial-temporal scene graphs (STSG)** for pixel-level grounding and adopt a **Video-of-Thought** multi-step reasoning pipeline.

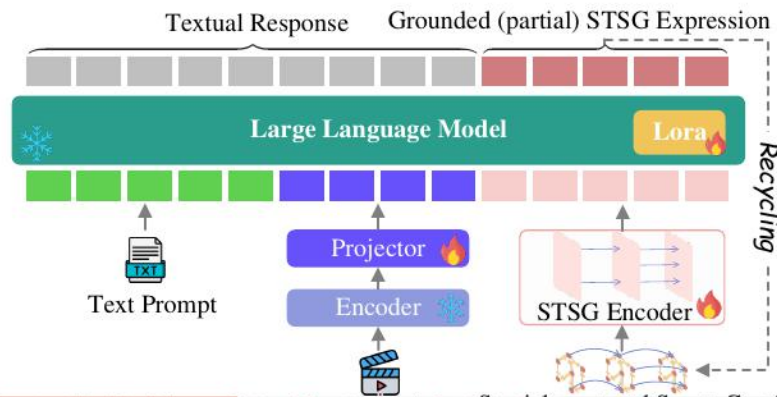
Question: What will happen to *the red oil tanker truck*?



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Method: MotionEpic Overview

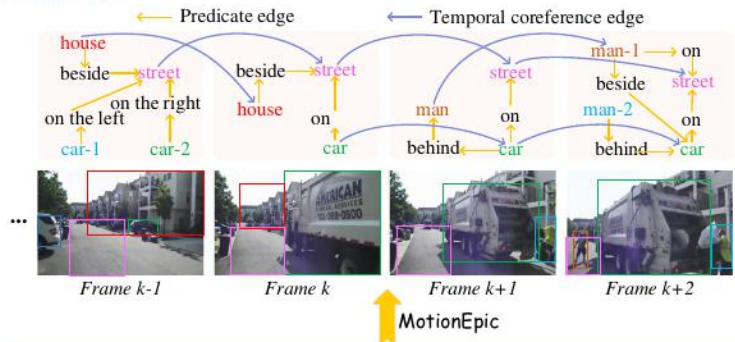
- Backbone: **Vicuna-7B (v1.5)** with ViT-L/14 encoder and **Q-Former** projector.
- STSG encoder: **Graph Transformer** with recurrent propagation for multi-frame graphs.
- Supports: **video + STSG** as input; **STSG generation** for grounding.



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Method: Integrating STSG Representation

- STSG: sequence of per-frame SGs with triplets (*subject*, *predicate*, *object*) and **temporal coreference edges**.
- Nodes: category c_i , feature f_i , bbox $b_i = (x, y, w, h)$.
- MotionEpic **understands and generates** STSGs, enabling **pixel-level grounding**.



Frame k-1: {
Objects: [{"car-1": [0.0,13.4,7.0,8.1]}, {"car-2": [27.0,12.9,9.0, 4.0]},
["house", [15.0, 0.0, 24.5, 17.0]], ["street", [10.0, 12.5, 16.5, 15.0]]],
Temporal Coreference Edges: [{"start": 0, "end": 1, "subject": "car-1", "object": "car-1", "predicate": "on the left"},
{"start": 0, "end": 1, "subject": "car-2", "object": "car-2", "predicate": "on the right"},
{"start": 0, "end": 1, "subject": "house", "object": "house", "predicate": "beside"},
{"start": 0, "end": 1, "subject": "street", "object": "street", "predicate": "beside"},
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{"start": 3, "end": 4, "subject": "car-2", "object": "car-2", "predicate": "on"}]}

Method: Grounding-aware Tuning

- Goal: **STSG-free inference** via strong video-STSG grounding.
- Coarse objectives:
 - $\mathcal{L}_1$: pairedness of video vs. STSG.
 - $\mathcal{L}_2$: generate full STSG from video.
- Fine objectives:
 - $\mathcal{L}_3$: tracklet(s) given video + action description.
 - $\mathcal{L}_4$: actions and tracklet(s) given key object(s).
 - $\mathcal{L}_5$: object label + tracklet from a bbox.
- Training: **instruction tuning** with LoRA; video pretraining on WebVid; then instruction tuning (VideoChat/Video-ChatGPT data).

Experimental Method: Video-of-Thought (VoT) Pipeline

- Decompose complex QA into **five chained steps**:
- 1) Target identification; 2) **Object tracking** via STSG; 3) **Action analysis** + commonsense; 4) **Answer ranking**; 5) **Verification** (grounding and cognition).

Step-1: Task Definition and Target Identification

Input

You are an expert ... answer a question based on the given video.
For the question, several candidate answers ...

Given the question: [What is the relationship between the white truck and this neighborhood? A. Transportation B. Buildings C. Clean Services D. Entertainment Facilities]

What are the possible targets of the  mainly mentioned or involved?

Output

The involved targets are [the white truck], [the neighborhood]

Step-2: Object Tracking

1. Provide the tracklet of involved [the neighborhood] and [the

Experimental Setting

- Tasks: **Video QA** (complex reasoning).
- Fine-tuning datasets: **VLEP**, **STAR** (Int./Seq./Pre./Fea.), **IntentQA**, **Social-IQ**, **Causal-VidQA**, **NExT-QA**.
- Zero-shot: **MSR-VTT**, **ActivityNet**.
- Grounding tuning: **Action Genome** (10K STSGs) + 350K **WebVid-10M** parsed STSGs.
- Implementation: Vicuna-7B, ViT-L/14, Q-Former, 6-layer GT for STSG, 8 fps sampling; training on **16x A100**.
- Baselines: InternVideo, LLaMA-VQA, VLAP, SeViLA, TranSTR, HiTeA, VideoChat family, Video-LLaVA.

Experimental Results: Main Comparison I

- VoT with MotionEpic **consistently** surpasses SoTA across complex QA datasets.
- STSG integration and VoT contribute large gains over vanilla CoT prompting.

Table: Results on VLEP and STAR, plus IntentQA and Social-IQ (as shown in Table 1).

Model	VLEP	STAR				IntentQA	Social-IQ	
		Int.	Seq.	Pre.	Fea.		2-Way	4-Way
VLAP	69.6	70.0	70.4	65.9	62.2	-	-	-
Video-LLaVA	66.3	67.7	63.8	75.9	-	-	-	-
Video-LLaVA+STSG	67.0	65.9	68.9	58.7	53.7	64.9	70.4	41.7
MotionEpic (CoT)	68.2	66.8	69.6	60.6	57.4	66.1	71.7	43.0
MotionEpic (VoT)	73.4	71.5	72.6	66.6	62.7	70.8	72.8	45.0

Experimental Results: Main Comparison II and Zero-shot

Table: Causal-VidQA results (as shown in Table 2).

Model	Acc@D	Acc@E	Acc@P			Acc@C		
			A	R	AR	A	R	AR
Video-LLaVA	73.7	74.4	67.6	65.4	47.7	68.0	64.9	51.5
MotionEpic (CoT)	78.5	77.2	70.1	70.8	52.4	71.2	69.1	55.0
MotionEpic (VoT)	81.2	83.0	74.3	73.7	54.7	74.5	73.8	58.6

Table: Zero-shot Video QA (as shown in Table 3).

Model	MSR-VTT	ActivityNet	NExT-QA	STAR	AVG
Video-LLaVA (CoT)	60.0	46.9	59.5	52.0	54.6
MotionEpic (CoT)	63.1	50.0	61.9	56.5	57.8
MotionEpic (VoT)	66.2	54.6	66.5	61.7	62.3

Ablation and Analysis

- MotionEpic shows strong **video grounding** (objects, triplets, actions) on Action Genome.
- Grounding-aware tuning matters: removing $\mathcal{L}_2$ or $\mathcal{L}_4$ causes the biggest drops.

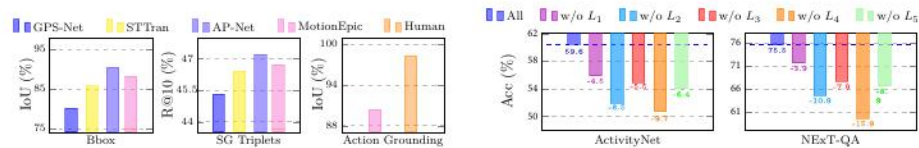
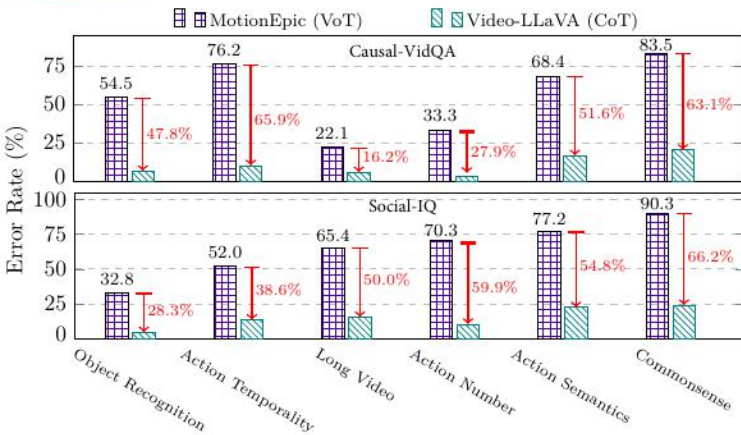


Figure: Left: Grounding performance (Fig. ??). Right: Zero-shot drops after ablations (Fig. ??).

Qualitative and Human Analysis

- **Human-level reasoning**: MotionEpic+VoT approaches human accuracy on hard subsets; reduces errors in **action semantics** and **commonsense**.



Deficiencies

- Requires **grounding-aware tuning** with video-STSG pairs; quality of parsed STSGs impacts performance.
- **Computational cost**: large models and multi-step VoT pipeline.
- Potential **commonsense biases** and error propagation across steps.
- Handling **very long videos** and dense events can be challenging at 8 fps.

Future Research

- More efficient **end-to-end** video-STSG grounding and **lightweight** VoT controllers.
- **Stronger verification** modules combining perception and cognition constraints.
- Extend beyond QA to **forecasting, planning, dialog, and real-time** applications.
- Safer deployment: **licensing, misuse prevention, and energy-aware** training.

Thank You

- Paper: Video-of-Thought (ICML 2024)
- Code/Project: <https://haofei.vip/VoT>
- Contact: Meishan Zhang (zhangmeishan@hit.edu.cn)